# QF256: A Fixed-Profile 256-Tone Modem with QRA Coding over GF(256)

Isak Paulo de Andrade Ruas
PY3KX
isakruas@gmail.com

October 2026

**Abstract**

QF256 combines a 256-tone, phase-continuous waveform with nonbinary repeat–accumulate coding for 77-bit messages. This article specifies the complete coding and receiver chain, including automatic acquisition over ±125 Hz and a fixed message-acceptance gate. The performance study uses the three mid-latitude multipath/fading profiles specified by ITU-R F.1487-0, followed by an ideal 300–2700 Hz audio frontend, with 128 frames at each of six noise levels per profile and two transmit-power constraints. The 2304 randomized cases are paired across the power settings, giving 4608 outcome records. Correct acceptance, erasures, and incorrect acceptance are reported with cell-level uncertainty; noise-only captures and acquisition-boundary cases are evaluated separately. Unit-average 50% acceptance crossings are approximately −19.3, −19.5 and −19.3 dB for quiet, moderate and disturbed; all three unit-peak crossings round to −18.7 dB. These are interpolations of a 2 dB grid, not sub-decibel resolution or a ranking of the profiles. The independent 4096-capture noise check accepts no messages, giving a one-sided 95% upper bound of $7.31 \times 10^{-4}$ per 15-second capture. SNR is referenced to transmit power and a 2500 Hz noise bandwidth, without renormalizing individual fades. The complete filtered burst lasts 9.733333 seconds. Waveform bandwidth, peak power, and the receiver's SNR-estimation error are measured alongside decoding outcomes. Source, per-trial records, statistical analysis, and figure-generation code accompany the manuscript.

## 1 Introduction

Short-message HF links must recover a small payload despite uncertain arrival time, carrier offset, and time-varying multipath. A noncoherent frequency receiver can supply a full vector of tone likelihoods to a nonbinary code, retaining information that a hard tone decision discards. Factor-graph inference provides the framework for combining those local observations with sparse coding constraints [1]. The QRA construction used here supplies the particular graph and metric machinery [11].

The practical object of study is the complete receiver, including acquisition and message acceptance. A decoder fed a known symbol boundary does not measure this behavior. Likewise, a quoted noise level is ambiguous unless its reference bandwidth, transmit-power constraint, and treatment of fading are specified. QF256 therefore fixes the waveform, coding map, noise units, blind search, and acceptance rule as one reproducible system.

The contribution is a parameter-complete 77-bit implementation and a statistical characterization under three explicitly defined HF channel profiles. The study reports correct acceptance, erasure, erroneous acceptance, noise-only acceptance, and waveform cost. The measurements evaluate a fixed receiver; the accompanying analysis regenerates the results

from individual records. Detailed command-line and library contracts are provided in the software documentation [8].

The payload is an opaque sequence of 77 bits whose interpretation is defined by the application. Table 1 lists the fixed parameters.

| Quantity | Fixed value |
|---|---|
| Source payload | 77 bits |
| Tone alphabet | 256 frequencies |
| Symbol duration $T_s$ | 0.128 s |
| Tone separation $\Delta f = 1/T_s$ | 7.8125 Hz |
| Complex sample rate $f_s$ | 12,000 Hz |
| Samples per symbol $N_s$ | 1536 |
| Transmitted code / pilot symbols | 60 / 16 |
| Unfiltered burst duration | $76T_s = 9.728$ s |
| Transmit FIR | 65 taps; 2000 Hz cutoff; Kaiser $\beta = 8.6$ |
| Filtered frame length $N_f$ | 116,800 complex samples |
| Active frame duration $T_f = N_f/f_s$ | 9.733333333333333 s |
| Default file capture | 180,000 samples; 15 s |
| Pilot-evidence threshold $\eta_0$ | 30.220758775848424 |

Table 1: The 12 kHz physical profile.

# 2 Payload packing and nonbinary coding

## 2.1 Bit order and CRC

Let $b_1, \ldots, b_{77} \in \{0, 1\}$ be the source bits in their public API order. The encoder packs them most significant bit first into ten bytes $m_1, \ldots, m_{10}$. The first nine bytes carry 72 source bits. The five remaining source bits occupy the high five bits of $m_{10}$; its low three bits are fixed to zero. Thus the CRC input is 80 bits: the 77 source bits followed by three public zero bits.

The core computes CRC-16/CCITT-FALSE over these ten bytes, processing each byte from its most significant bit to its least significant bit. The polynomial is `0x1021`, the initial register is `0xFFFF`, and there is no reflection or final XOR. The high and low CRC bytes occupy $m_{14}$ and $m_{15}$, respectively. The remaining information positions are fixed as

$$m_{11} = m_{12} = m_{13} = 0. \tag{1}$$

## 2.2 Accumulator construction and transmitted positions

The mother construction has 65 field-valued variables: 15 information positions followed by 50 accumulator positions. Arithmetic uses GF(256) with primitive polynomial `0x11D`; field addition is XOR. Writing the accumulator variables as $a_1, \ldots, a_{50}$, the encoder uses

$$a_0 = 0,$$
$$a_i = a_{i-1} \oplus h_i m_{v_i}, \qquad 1 \le i \le 50, \tag{2}$$
$$0 = a_{50} \oplus h_{51} m_{v_{51}}. \tag{3}$$

The fixed graph has 51 check nodes and 151 edges. Its coefficients satisfy the termination identity in (3). Appendix A records the graph indices and coefficient construction used by the source.

The following 60 variables select data tones, in this order:

$$(m_1, \ldots, m_{10}, a_1, \ldots, a_{50}). \tag{4}$$

The three zero information symbols in (1) are *shortened*: the decoder knows their values. The two CRC symbols $m_{14}, m_{15}$ are *punctured*: they are not transmitted, and their initial decoder priors are uniform. Their values must be recovered through the code constraints before the CRC comparison can pass. The low three bits of $m_{10}$ are also constrained to zero in the decoder prior.

### 2.3 The operational codebook

The mother graph is linear over GF(256): multiplication by a fixed coefficient and XOR preserve field addition. The operational encoder applies additional bit and CRC constraints before selecting the transmitted positions. Let $F : \{0,1\}^{77} \to \mathrm{GF}(256)^{60}$ denote that complete map. The first ten transmitted symbols retain every input bit, so $F$ is injective and its image contains exactly $2^{77}$ words.

This image is not a generic linear code over GF(256). The low-three-bit restriction is not closed under arbitrary field multiplication, and the CRC with initial register `FFFF` is affine in its input bits. In the polynomial bit representation, fixed field multiplication, packing, and puncturing are binary-linear operations; their composition with this CRC makes $F$ a binary-affine map. Cardinality and injectivity do not specify its minimum distance. The ratio $77/(60 \cdot 8) = 77/480$ counts source bits per transmitted binary-label position, excluding pilots; it is not a field-symbol dimension.

The graph and coefficients are adapted from the attributed QRA source material. The GF(256) construction uses the numeric coefficients specified in Appendix A [7, 11].

## 3 Frame allocation and transmit waveform

### 3.1 Public pilots and tone mapping

Number frame symbols from 1 through 76. The sixteen pilots occupy

$$p_j = 1 + 5(j-1), \qquad j = 1, \ldots, 16. \tag{5}$$

The remaining positions carry the 60 values in (4). The layout is fifteen groups of one pilot followed by four data symbols, then one final pilot, as illustrated in Figure 1.

The pilot tone sequence is deterministic and public. Starting with $r_0 = 167$, it is generated by

$$r_j = (109 r_{j-1} + 89) \bmod 256, \qquad j = 1, \ldots, 16. \tag{6}$$

For any pilot or data value $m \in \{0, \ldots, 255\}$, the baseband frequency is

$$f_m = (m - 127.5)\Delta f, \qquad \Delta f = 7.8125\,\mathrm{Hz}. \tag{7}$$

The extreme tone centers are $-996.09375\,\mathrm{Hz}$ and $+996.09375\,\mathrm{Hz}$. Their span is $1992.1875\,\mathrm{Hz}$; the convention $256\Delta f$ gives a nominal 2 kHz tone lattice. Filtering and finite burst duration also determine the actual occupied spectrum.

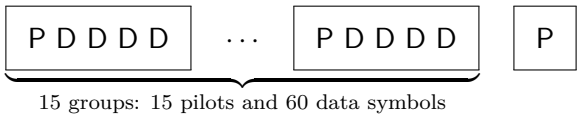



Figure 1: The 76-symbol frame. P is a public pilot and D is a transmitted code symbol. Complete filter tails extend the output beyond the symbol train.

### 3.2 Phase continuity, filtering, and normalization

Before filtering, each symbol is a unit-magnitude complex sinusoid. Since $f_m T_s = m - 127.5$, every tone advances the phase by an odd multiple of $\pi$ during one symbol. The boundary phasor therefore alternates between $+1$ and $-1$ independently of the data. The implementation uses that property to generate a continuous-phase sequence at tone transitions.

The finite symbol train is convolved with a real, symmetric 65-tap low-pass kernel. At the public sample rate, let $q = -32, \ldots, 32$, $f_c = 2000\,\text{Hz}$, and $\beta = 8.6$. Before DC-gain normalization, its samples are

$$\widetilde{h}[q] = \begin{cases} 2f_c/f_s, & q = 0, \\ \sin(2\pi f_c q/f_s)/(\pi q), & q \neq 0 \end{cases} \frac{I_0\left(\beta\sqrt{1-(q/32)^2}\right)}{I_0(\beta)}, \tag{8}$$

with $h[q] = \widetilde{h}[q]/\sum_{u=-32}^{32} \widetilde{h}[u]$. Here $I_0$ is the modified Bessel function. The causal stored kernel has 32 samples of group delay. Full convolution is retained: the 116,736-sample symbol train gains 64 samples, producing all 116,800 output samples without cropping either tail.

The 2000 Hz value is the one-sided cutoff parameter of this transmit FIR, not the occupied bandwidth of the modem. The tone-center span is 1992.1875 Hz, and occupied bandwidth is measured separately from the waveform's energy spectrum. Neither quantity is the receiver frontend's 300–2700 Hz passband.

If $z[n]$ is this filtered frame, the transmitter returns

$$x[n] = \frac{z[n]}{\sqrt{N_f^{-1} \sum_{k=0}^{N_f-1} |z[k]|^2}}. \tag{9}$$

Consequently the mean complex power of the complete active frame is one in the core's floating-point representation. Filtering introduces instantaneous envelope variation [6].

## 4 Receiver processing

The receiver accepts complex IQ at the fixed sample rate, estimates noise from the supplied capture, determines time and frequency offsets, measures tone energies, and decodes one codeword. It returns at most one message per capture. Figure 2 summarizes the sequence.

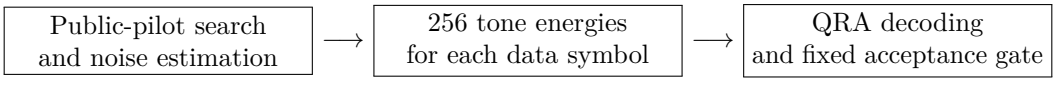



Figure 2: Receiver processing. The caller supplies the IQ capture and uses automatic acquisition. Aligned-input decoding is also available for tests.

### 4.1 Noise estimate and units

For the sample-domain convention used here,

$$\mathbb{E}\{|w[n]|^2\} = f_s N_0. \tag{10}$$

The receiver divides the input into complete, nonoverlapping 512-sample blocks. Each block is multiplied by the periodic Hann window $u[n] = \frac{1}{2} - \frac{1}{2}\cos(2\pi n/512)$ and transformed with an unnormalized FFT. Positive and negative bins in the public 2500–5000 Hz guard bands are pooled; at 12 kHz these correspond to signed bin indices $\mathcal{K} = \{\pm 107, \ldots, \pm 213\}$.

If $X_b[k]$ denotes a windowed block spectrum, $B$ is the number of complete blocks, and $W = \sum_{n=0}^{511} u[n]^2$, the estimate is

$$\widehat{N}_{0,g} = \max\left[\frac{\sum_{b=1}^{B}\sum_{k\in\mathcal{K}}|X_b[k]|^2}{B|\mathcal{K}|W f_s},\ 10^{-20}\right]. \tag{11}$$

This guard estimate assumes representative noise in those bands. An audio frontend can remove the guards while retaining the modem tones. The receiver therefore also estimates in-band noise from the acquired data-symbol projections. For symbol $i$, let $E_{(k),i}$ be the $k$th ordered energy among its 256 candidate tones. It computes

$$\widehat{N}_{0,t} = \max\left[\frac{1}{60c}\sum_{i=1}^{60}\frac{E_{(128),i} + E_{(129),i}}{2},\ 10^{-20}\right], \tag{12}$$

$$\widehat{N}_0 = \begin{cases} \widehat{N}_{0,t}, & \widehat{N}_{0,g} < \widehat{N}_{0,t}/2, \\ \widehat{N}_{0,g}, & \text{otherwise.} \end{cases} \tag{13}$$

The constant $c = 0.6951041202281072$ is the expected mean of order statistics 128 and 129 from 256 independent unit-mean exponential variables. This normalization is exact for that noise model, not a claim that acquired, filtered signal projections are independent exponentials. The median limits sensitivity to an occupied tone but is not immune to colored interference or widespread signal leakage. The public result field `noise_variance` is $f_s\widehat{N}_0$, an estimate of the equivalent prefilter complex white-noise variance; it is not the total remaining sample variance after band limitation.

### 4.2 Time and frequency acquisition

For a candidate zero-based burst offset $s$ and residual frequency $\nu$, the pilot score is

$$S(s,\nu) = \frac{1}{f_s N_s}\sum_{j=1}^{16}\left|\sum_{n=0}^{N_s-1} y[s + 32 + (p_j - 1)N_s + n]\, e^{-\mathrm{j}2\pi(f_{r_j}+\nu)n/f_s}\right|^2. \tag{14}$$

The 32-sample offset aligns the rectangular tone integration with the transmit filter's group delay.

Automatic acquisition searches starts from 0.3 through 0.7 seconds and residual CFO from −125 through +125 Hz. It maximizes the pilot score over a coarse grid, then two local grids around the preceding winner:

| Stage | Time step | CFO step (Hz) | Time region | CFO region |
|---|---|---|---|---|
| Coarse | 96 samples | 3.90625 | Full interval | Full interval |
| Middle | 16 samples | 0.244140625 | ±96 samples | ±3.90625 Hz |
| Fine | 4 samples | 0.0244140625 | ±16 samples | ±0.244140625 Hz |

All local regions remain clipped to the public bounds. The coarse frequency grid includes both endpoints exactly. The final grid resolution is not an accuracy guarantee under noise or fading. The receiver evaluates one winning acquisition hypothesis for decoding, rather than trying every grid point through the FEC decoder.

The diagnostic `known` setting evaluates only $s = 0, \nu = 0$; it requires an already aligned frame and is not an alternative automatic mode. At least 125,200 samples are required for automatic acquisition, enough to include the filtered frame at the latest searched start. Both interfaces limit captures to 180,000 samples.

### 4.3 Tone energies and soft metrics

At the selected offset and CFO, the receiver measures the 256 candidate tone energies in each of the 60 data-symbol intervals, with the same $1/(f_s N_s)$ normalization as (14). The implementation uses three 512-point FFT lanes to evaluate the required bins of the 1536-sample transform. For unfiltered white noise under the convention in (10), this projection has complex noise variance $N_0$. Finite-band filtering and fading need not preserve the independent exponential-energy model exactly; the experiments include the specified frontend instead of assuming it has no effect.

The core uses a fixed noncoherent intrinsic metric. Let $E_{m,i}$ be the energy for tone $m$ in data interval $i$, and define

$$\gamma = 6\left(\frac{13}{63}\right)10^{0.28}, \qquad z_{m,i} = 2\sqrt{\gamma E_{m,i}/\widehat{N}_0}, \tag{15}$$

$$L(z) = \frac{z^2(z + 0.039)}{0.9931z^2 + 2.6936z + 0.5185}. \tag{16}$$

The rational expression approximates $\log I_0(z)$ and is part of the attributed QRA metric implementation. For each interval, $L(z)$ is shifted by its maximum, floored at $-700$ in the exponent, exponentiated, and normalized over the tone alphabet. These probabilities are the decoder observations [7, 11].

### 4.4 Iterative decoding and final acceptance

The decoder performs sum–product updates over the fixed nonbinary graph, using Walsh–Hadamard transforms to evaluate additive-field convolutions [1, 7]. It runs at most 100 iterations [7]. The transmitted symbols receive their measured priors; the known zero positions and the three low zero bits of $m_{10}$ are constrained as described in Section 2.

Before declaring a raw successful decode, the implementation requires an extrinsic-convergence score of at least $65 - 0.01$, zero graph syndrome, and agreement of the recomputed CRC bytes with the recovered punctured CRC positions. The CRC is a final check, not an additional factor enforced during message passing. The convergence score is the sum, over all 65 variables, of each normalized maximum extrinsic probability.

The application acceptance test adds a pilot gate:

$$\eta = S(\widehat{s}, \widehat{\nu})/\widehat{N}_0 - 16, \tag{17}$$

$$\text{accepted} \iff \text{crc_ok} \ \wedge \ (\eta \geq \eta_0), \qquad \eta_0 = 30.220758775848424. \tag{18}$$

Here `crc_ok` denotes the complete raw core success condition, including convergence and syndrome checks. Both file frontends use the same threshold $\eta_0$.

### 4.5 CRC-conditioned SNR estimate

For a CRC-valid decode, let $\widehat{t}_i$ be the recovered tone in data interval $i$ and $E_{\widehat{t}_i,i}$ its received matched-projection energy. The receiver regenerates a clean complete frame from that payload, normalized to unit mean complex power by (9), and measures the corresponding correct-bin energies $E^{(1)}_{\widehat{t}_i,i}$. The estimated received active-frame power and reported SNR are

$$\widehat{P} = \frac{\sum_{i=1}^{60}(E_{\widehat{t}_i,i} - \widehat{N}_0)}{\sum_{i=1}^{60} E^{(1)}_{\widehat{t}_i,i}}, \tag{19}$$

$$\widehat{\mathrm{SNR}}_{2500} = 10\log_{10}\left(\frac{\widehat{P}}{2500\widehat{N}_0}\right). \tag{20}$$

In aligned, unfiltered static AWGN with true $N_0$, a correct-bin energy has expectation $PE^{(1)}_{\widehat{t}_i,i} + N_0$. Subtracting one $\widehat{N}_0$ per interval therefore estimates received signal power without treating the pilot evidence as SNR. This identity does not establish an unbiased operational estimate: the receiver also estimates synchronization, noise, and payload, and filtering or fading can change the tone projections. The public validity flag is set only when the CRC is valid and the noise, reference sum, signal sum, and final logarithm are finite with positive power. No Doppler or SNDR estimate is reported.

## 5 Performance-study method

### 5.1 Channel and observation model

The study uses the quiet, moderate and disturbed mid-latitude profiles specified by Recommendation ITU-R F.1487-0, Annex 3, Sections 3.1–3.3 [3], with Gaussian-scatter tapped-delay fading [2]. Their finite spectral implementation is defined here. Let $x[n]$ be the normalized transmit frame, extended by zero outside its support, and let $A$ apply the chosen transmit-power constraint. With $t = (n-s)/f_s$, the observation before frontend filtering is

$$r[n] = Ae^{\mathrm{j}2\pi\nu t}\sum_{\ell=1}^{2}\sqrt{P_\ell}\, g_\ell(t)x[n-s-d_\ell] + w[n]. \tag{21}$$

Here $s$ is an integer start offset, $\nu$ is constant CFO, and the path processes are mutually independent. Each has the finite spectral representation

$$g_\ell(t) = \sum_{k=1}^{64} c_{\ell k}e^{\mathrm{j}2\pi F_{\ell k}t}, \quad c_{\ell k} \sim \mathcal{CN}(0, 1/64), \quad F_{\ell k} \sim \mathcal{N}(0, (B_D/2)^2). \tag{22}$$

Averaging over both coefficients and frequencies gives $\mathbb{E}|g_\ell|^2 = 1$ and correlation $R_g(\tau) = \exp(-\pi^2 B_D^2\tau^2/2)$. A realization contains 64 spectral lines; it approximates the continuous Gaussian spectrum. Individual fades are not power-normalized. Here the Doppler spread $B_D$ is twice the Gaussian spectrum's standard deviation, as in the recommendation, not its full width at half maximum. Each path's mean Doppler is zero, as specified in Annex 3. The common $\nu \in [-2, 2)$ Hz is an additional tuning offset for this receiver experiment, not Doppler spread or a parameter of the ITU profile.

The delays in Table 2 are exact integer sample shifts. Fading is evaluated from the frame origin, and delayed tails are retained. The start is drawn uniformly from $[0.4, 0.6)$ s and rounded to the nearest sample; CFO is uniform over $[-2, 2)$ Hz. There is no frequency drift.

| Profile | Delays | Path powers | $B_D = 2\sigma_F$ |
|---|---|---|---|
| Quiet | 0, 0.5 ms (0, 6 samples) | $1/2, 1/2$ | 0.1 Hz |
| Moderate | 0, 1 ms (0, 12 samples) | $1/2, 1/2$ | 0.5 Hz |
| Disturbed | 0, 2 ms (0, 24 samples) | $1/2, 1/2$ | 1 Hz |

Table 2: The three ITU-R F.1487-0 mid-latitude profiles at $f_s = 12$ kHz. Both independently fading paths have equal ensemble power.

The receiver receives only the capture and fixed public configuration; it searches the larger region specified in Section 4. Each cell measures frame acceptance over 128 frames.

The receiver input is $y = \mathcal{H}r$, where $\mathcal{H}$ is an ideal FFT-domain mask retaining $|f| \leq 1200$ Hz. This models a 300–2700 Hz USB audio channel translated about 1500 Hz. The mask acts on signal and noise together. For each 180,000-sample observation, the simulator generates a 524,288-sample context, places the delayed signal in its central observation window, filters the complete context, and retains the central observation. Noise extends beyond both observation boundaries; each boundary is 172,144 samples from the periodic FFT seam. The model excludes physical filter ripple and transition bands. It preserves finite-context FFT behavior rather than asserting an exact infinite-duration brick-wall realization.

Independent Gaussian noise is generated throughout that context before filtering, including samples outside the burst. For a reference level $\rho$ in dB,

$$N_0 = \frac{1}{2500\, 10^{\rho/10}}, \qquad w[n] \sim \mathcal{CN}(0, f_s N_0). \tag{23}$$

Each prefilter real quadrature has variance $f_s N_0/2$. Band limitation correlates received samples and reduces their total variance; the spectral density in the retained passband remains the reference $N_0$. Noise is not rescaled by the instantaneous faded receive power or by waveform bandwidth.

### 5.2 Transmit power, energy, and waveform measurements

In the unit-average setting $A = 1$, so the active-frame transmit power $P_{\text{av}} = 1$ and energy $E = \sum |Ax[n]|^2/f_s = T_f$. In the unit-peak setting, $A = 1/\sqrt{P_{\text{pk,ref}}}$, where $P_{\text{pk,ref}}$ is the larger of the peaks obtained by regenerating the waveform at 48 and 192 kHz, normalized by the 12 kHz frame's pre-normalization mean power. The 48 and 192 kHz FIRs contain 257 and 1025 taps, respectively, preserving the kernel support of $\pm 32/12000$ s in (8) [6]. The check requires the two peak estimates to agree within 0.05 dB. Consequently, $P_{\text{av}} = 1/P_{\text{pk,ref}}$ and $E = T_f/P_{\text{pk,ref}}$ in that setting. These are measured oversampled peaks, not a proof of a continuous-time supremum.

The horizontal coordinate $\rho$ always denotes the unit-reference level in (23). Actual average transmit SNR and source-bit energy are

$$\text{SNR}_{2500} = \rho + 10\log_{10} P_{\text{av}}, \qquad (E_b/N_0)_{77} = \rho + 10\log_{10}(2500E/77). \tag{24}$$

The first equals $\rho$ only under unit-average normalization. These quantities refer to transmit power and ensemble channel normalization, not to the power of a particular faded receive frame or the power remaining after frontend filtering. Dividing by 77 counts the uniformly sampled message bits.

Bandwidth is measured on the complete active waveform before the channel. An unwindowed FFT is zero-padded to the next power of two at least $4N_f$ (524,288 points here). The ordered, two-sided squared spectrum defines cumulative-energy quantiles $f_{0.005}$ and

$f_{0.995}$, interpolated within bins. The reported width is $B_{99} = f_{0.995} - f_{0.005}$; FFT spacing is approximately 0.0229 Hz. PAPR is $10\log_{10}(P_{\mathrm{pk,ref}}/P_{\mathrm{av,12k}})$ for the unscaled waveform. Here $P_{\mathrm{av,12k}}$ is the mean power of the normalized 12 kHz frame, approximately one, before unit-peak scaling. Both width and PAPR are invariant under the subsequent constant peak scaling.

### 5.3 Sampling and fixed receiver configuration

For each profile, the reference levels are $\{-24, -22, -20, -18, -16, -14\}$ dB, with 128 separately seeded frames in every cell. The two power settings reuse the payload, fading, noise, start, and CFO draws: there are 2304 underlying randomized cases and 4608 outcome records across 36 cells, not 4608 independent realizations. Outcomes within a cell use different payload, channel, and noise seeds. The analysis makes no independence assumption between power settings.

The message generator samples all 77 input bits: one uniform draw supplies one bit and nineteen further draws supply four bits each. The resulting 20-digit hexadecimal representation includes the three leading padding bits required by the file interface, not fixed payload bits. The next two draws set start and CFO. With profile index $q \in \{2, 3, 4\}$ for quiet, moderate and disturbed, SNR index $j \in \{1, \ldots, 6\}$, trial $i \in \{1, \ldots, 128\}$, and $c = 6(q-1) + j$, the seeds are

$$\begin{aligned} s_p &= 20271011 + 100003c + 101i, \\ s_h &= 120271011 + 200003c + 307i, \\ s_w &= 220271011 + 300007c + 911i + 29. \end{aligned} \tag{25}$$

The Fortran random generator is initialized separately for these streams; Gaussian values use the documented Box–Muller transform. Bitwise replay therefore depends on the retained GNU Fortran runtime convention. The standalone harness implements the same initialization and draw order.

The receiver configuration, including $\eta_0 = 30.220758775848424$, is a fixed design input to this campaign. The study does not derive or tune that threshold from its observed outcomes. A separate 2048-capture noise diagnostic lot describes evidence quantiles and raw/gated outcomes; its 99th percentile must not be substituted for the operating threshold. Source and per-trial records identify the exact implementation and configuration [9].

### 5.4 Endpoints and uncertainty

A signal trial succeeds only when the final acceptance rule passes and all 77 recovered bits equal the transmitted bits. An erasure is a trial with no accepted message; an incorrect acceptance passes the rule with different bits. These three counts sum to the cell size. Frame error rate is one minus correct acceptance, so it includes both erasures and incorrect acceptances.

For $k$ correct acceptances among $n = 128$ trials, $\widehat{p} = k/n$. The pointwise two-sided 95% Wilson interval [4] is

$$\frac{\widehat{p} + z^2/(2n) \;\pm\; z\sqrt{\widehat{p}(1-\widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \qquad z = 1.95996398454. \tag{26}$$

These are cell intervals, not simultaneous confidence bands. The grid average weights all eighteen profile/SNR cells equally and is reported as a descriptive summary, without treating heterogeneous cells as one binomial population.

The estimated 50% crossing linearly interpolates the two adjacent empirical rates that bracket 0.5. It is reported to 0.1 dB because the grid spacing is 2 dB. To express sampling uncertainty without assuming a subgrid curve shape, each curve also receives a conservative grid bracket. Exact binomial intervals use a Bonferroni allocation across its six points (tail probability 0.05/12). The bracket extends from the highest grid point whose upper bound is below 0.5 to the lowest whose lower bound exceeds 0.5. Its confidence interpretation requires monotone success probability in SNR; it is not an interval for the linear-interpolation model.

The independent noise-only check uses 4096 fresh 15-second captures and retains the same blind receiver. Reference levels cycle through the six noise scales in (23); there is no received SNR without a signal. Diagnostic and check seeds are disjoint. They use bases 40271011 and 60271011, respectively. For global index $g$ starting at one, set $j = 1 + ((g-1) \bmod 6)$, $i = 1 + \lfloor (g-1)/6 \rfloor$, and $s_w = b + 300007j + 911i + 29$. Both lots use the same ideal frontend as the signal study. For zero accepted messages, the one-sided exact 95% binomial upper bound is [5]

$$p_U = 1 - 0.05^{1/4096} = 0.0007311125565. \tag{27}$$

It is exact under a common capture probability. If probabilities differ between noise scales, it is conservative for their allocation-weighted mean: for independent captures, $\prod_i (1-p_i) \le (1-\overline{p})^{4096}$. Diagnostic samples are excluded from this denominator. Separate indexed seed ranges do not prove the statistical independence of a pseudorandom generator; the model treats the separately initialized streams as independent. The deliberately paired power settings are not independent trials.

### 5.5 Acquisition geometry and estimator checks

A separate AWGN experiment exercises the full acquisition limits through the same nominal audio passband. With audio centers 1475 and 1525 Hz, the ideal complex masks are $[-1175, 1225]$ Hz and $[-1225, 1175]$ Hz. Combining these centers with CFO limits $\pm 125$ Hz places the most extreme tone centers at 353.90625 and 2646.09375 Hz, 53.90625 Hz inside the corresponding audio edge. This is a tone-center geometry statement, not a guarantee that all waveform energy is inside the passband or that a practical filter has adequate edge gain.

The signal check uses 128 payload/noise cases at each of $-20$, $-18$, and $-4$ dB, with 125,200-sample observations. The first eight cases at each level cover every Cartesian combination of start 3600/8400 samples, CFO $-125/+125$ Hz, and frontend center 1475/1525 Hz. Other starts are uniform integers in the same bounds, CFO is uniform over the frequency interval, phase is uniform on $[-\pi, \pi)$, and frontend centers alternate. An independently seeded 4096-capture noise lot alternates those two masks. These trials measure boundary acquisition separately from the main fading study's $\pm 2$ Hz CFO distribution; the two sets are not pooled. They call the public float32 C interface in separate processes. Signal filtering uses a 262,144-sample context and retains its central 125,200 samples. Per-case seeds, numerical-library versions, and source identities accompany the acquisition records [9].

The estimator check uses one deterministic 77-bit payload and 128 cases per point. In AWGN it measures known aligned transmitted tones at six levels from $-24$ through $-14$ dB, using the receiver's selected noise estimate. Fading cases use automatic acquisition, a start of 0.5 s, CFO 3.2 Hz, and levels $-18$, $-16$, and $-14$ dB. Their SNR errors are recorded only for correct accepted messages with valid estimates. The six AWGN rows and three rows for each fading profile give 15 rows and 1920 attempted estimates, separately from the main signal grid. Both use the centered ideal frontend. Thus AWGN and fading estimator rows have different conditioning and are not direct sensitivity comparisons.

# 6 Results

## 6.1 Correct-message acceptance and erasures

Figure 3 and Tables 3 and 4 separate the two power constraints while retaining all three profiles and six reference levels. Their confidence intervals are pointwise, not simultaneous bands. The equal-weight grid summary has 1232 correct acceptances in 2304 records (53.5%) under unit-average power and 1088/2304 (47.2%) under unit-peak power. For quiet, moderate and disturbed respectively, the unit-average totals are 407/768, 417/768 and 408/768; the unit-peak totals are 365/768, 364/768 and 359/768. These fractions summarize the deliberately balanced grid, not an operating distribution of HF conditions.

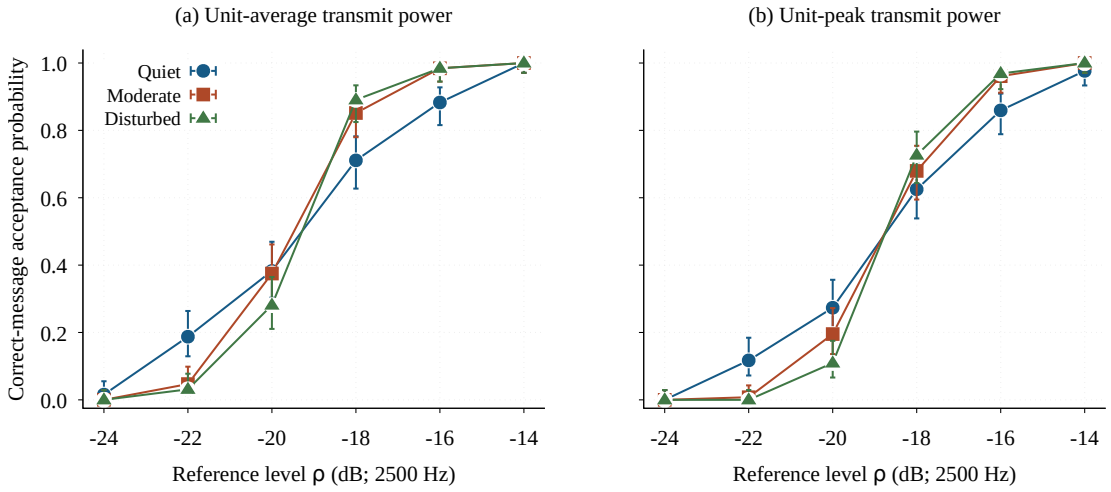



Figure 3: Correct-message acceptance with pointwise 95% Wilson intervals; 128 frames per point. Lines connect measured points. In (a), the reference level $\rho$ equals average transmit SNR in 2500 Hz. In (b), average transmit SNR is approximately 0.711 dB lower because of peak normalization. The two panels reuse the same randomized cases.

At $-24$ dB, quiet fading under unit-average power has two correct acceptances; the other five conditions have none. At $-14$ dB, five conditions accept all 128 messages and quiet fading under unit-peak power accepts 125/128. Even a 128/128 cell has a Wilson interval of approximately $[0.971, 1]$, not proof of error-free operation. No incorrect message is accepted in any of the 4608 signal records; the other outcomes are erasures. Because the power settings are paired, these are not 4608 independent opportunities for an undetected error.

## 6.2 The 50% acceptance crossing

All six crossings interpolate between the measured $-20$ and $-18$ dB points. At unit-average power they are approximately $-19.3$, $-19.5$ and $-19.3$ dB for quiet, moderate and disturbed. All three unit-peak values round to $-18.7$ dB. The quiet unit-average grid bracket is $[-22, -18]$ dB; the other five are $[-20, -18]$ dB. These overlapping brackets and the 2 dB measurement spacing do not support a ranking based on the small interpolated differences.

The observed curves also change order across the grid. For example, unit-average quiet and disturbed have 24 and 4 correct acceptances at $-22$ dB, but 91 and 114 at $-18$ dB. The ITU profile names identify delay/Doppler parameters rather than a universal ordering of this receiver's frame-acceptance probability. The experiment changes both parameters and does not isolate a cause for the observed differences.

| Profile | Reference $\rho$ | $k/128$ | Rate | 95% CI |
|---|---|---|---|---|
| Quiet | −24 dB | 2/128 | 0.016 | [0.004, 0.055] |
| Quiet | −22 dB | 24/128 | 0.188 | [0.129, 0.264] |
| Quiet | −20 dB | 49/128 | 0.383 | [0.303, 0.469] |
| Quiet | −18 dB | 91/128 | 0.711 | [0.627, 0.782] |
| Quiet | −16 dB | 113/128 | 0.883 | [0.816, 0.928] |
| Quiet | −14 dB | 128/128 | 1.000 | [0.971, 1.000] |
| Moderate | −24 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Moderate | −22 dB | 6/128 | 0.047 | [0.022, 0.098] |
| Moderate | −20 dB | 48/128 | 0.375 | [0.296, 0.461] |
| Moderate | −18 dB | 109/128 | 0.852 | [0.780, 0.903] |
| Moderate | −16 dB | 126/128 | 0.984 | [0.945, 0.996] |
| Moderate | −14 dB | 128/128 | 1.000 | [0.971, 1.000] |
| Disturbed | −24 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Disturbed | −22 dB | 4/128 | 0.031 | [0.012, 0.078] |
| Disturbed | −20 dB | 36/128 | 0.281 | [0.211, 0.365] |
| Disturbed | −18 dB | 114/128 | 0.891 | [0.825, 0.934] |
| Disturbed | −16 dB | 126/128 | 0.984 | [0.945, 0.996] |
| Disturbed | −14 dB | 128/128 | 1.000 | [0.971, 1.000] |

Table 3: Unit-average power: correct acceptances and pointwise 95% Wilson intervals, with 128 frames per cell. The reference level $\rho$ equals average transmit SNR in 2500 Hz. Erasure and incorrect-acceptance counts and intervals are supplied separately in `study/cells.csv`.

| Profile | Reference $\rho$ | $k/128$ | Rate | 95% CI |
|---|---|---|---|---|
| Quiet | −24 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Quiet | −22 dB | 15/128 | 0.117 | [0.072, 0.184] |
| Quiet | −20 dB | 35/128 | 0.273 | [0.204, 0.356] |
| Quiet | −18 dB | 80/128 | 0.625 | [0.539, 0.704] |
| Quiet | −16 dB | 110/128 | 0.859 | [0.789, 0.909] |
| Quiet | −14 dB | 125/128 | 0.977 | [0.933, 0.992] |
| Moderate | −24 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Moderate | −22 dB | 1/128 | 0.008 | [0.001, 0.043] |
| Moderate | −20 dB | 25/128 | 0.195 | [0.136, 0.272] |
| Moderate | −18 dB | 87/128 | 0.680 | [0.595, 0.754] |
| Moderate | −16 dB | 123/128 | 0.961 | [0.912, 0.983] |
| Moderate | −14 dB | 128/128 | 1.000 | [0.971, 1.000] |
| Disturbed | −24 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Disturbed | −22 dB | 0/128 | 0.000 | [0.000, 0.029] |
| Disturbed | −20 dB | 14/128 | 0.109 | [0.066, 0.175] |
| Disturbed | −18 dB | 93/128 | 0.727 | [0.644, 0.796] |
| Disturbed | −16 dB | 124/128 | 0.969 | [0.922, 0.988] |
| Disturbed | −14 dB | 128/128 | 1.000 | [0.971, 1.000] |

Table 4: Unit-peak power: correct acceptances and pointwise 95% Wilson intervals, with 128 frames per cell. The reference level $\rho$ uses unit power; actual average transmit SNR follows (24). These cases share payload, channel and noise draws with Table 3; they are not independent repetitions.

| Profile | Power setting | 50% crossing | 95% grid bracket |
|---|---|---|---|
| Quiet | Average | $-19.3\,\mathrm{dB}$ | $[-22, -18]\,\mathrm{dB}$ |
| Moderate | Average | $-19.5\,\mathrm{dB}$ | $[-20, -18]\,\mathrm{dB}$ |
| Disturbed | Average | $-19.3\,\mathrm{dB}$ | $[-20, -18]\,\mathrm{dB}$ |
| Quiet | Peak | $-18.7\,\mathrm{dB}$ | $[-20, -18]\,\mathrm{dB}$ |
| Moderate | Peak | $-18.7\,\mathrm{dB}$ | $[-20, -18]\,\mathrm{dB}$ |
| Disturbed | Peak | $-18.7\,\mathrm{dB}$ | $[-20, -18]\,\mathrm{dB}$ |

Table 5: Interpolated 50% crossing in reference-level dB and conservative 95% grid brackets under a monotone-response assumption. Each bracket uses six simultaneous exact binomial intervals for its own curve; coverage is not joint across the six curves.

### 6.3 Noise-only acceptance

| Lot | Captures | Raw FEC accepts | Final accepts | Gate exceedances |
|---|---|---|---|---|
| Diagnostic | 2048 | 0 | 0 | 3 |
| Independent check | 4096 | 0 | 0 | 4 |

Table 6: Noise-only outcomes. Evidence-gate exceedance alone is not acceptance; the receiver also requires raw decoder success.

The independently seeded check has zero false accepts in 4096 captures, giving the upper bound in (27), about 0.0731% per 15-second capture. The six reference scales have counts 683, 683, 683, 683, 682 and 682 in increasing $\rho$ order. The pooled bound does not provide an equally small 95% upper limit at each scale. Neither lot has any raw FEC acceptance, so these records do not measure an observed reduction caused by gating. The diagnostic evidence 99th percentile is 26.4951, below the fixed gate 30.220758775848424; this descriptive quantile does not set or recalibrate the gate.

### 6.4 Acquisition-boundary checks

| SNR (dB) | Cases | Raw correct | Accepted correct | Exact corners |
|---|---|---|---|---|
| $-20$ | 128 | 116 | 115 | 8/8 |
| $-18$ | 128 | 128 | 128 | 8/8 |
| $-4$ | 128 | 128 | 128 | 8/8 |

Table 7: Independent AWGN acquisition checks over the complete timing/CFO interval and the two limiting frontend centers. Exact corners count correct messages that also pass the fixed gate.

All 24 sampled exact Cartesian corners succeed. At $-20\,\mathrm{dB}$, one raw correct decode is rejected by the evidence gate; the other unsuccessful cases are raw-decoder failures. No incorrect message is accepted. These results exercise the stated ideal-frontend geometry, not every possible noise or fading realization at a boundary.

The separate noise check with alternating frontend centers has zero raw and zero accepted decodes in 4096 captures, with two evidence-gate exceedances. Its own one-sided 95% upper acceptance bound is $7.31 \times 10^{-4}$ per capture. It is not pooled with the centered frontend check in Table 6, because the input ensembles differ.

### 6.5 SNR-estimation error

| Channel | Target (dB) | Valid / 128 | Bias (dB) | RMSE (dB) |
|---|---|---|---|---|
| AWGN | −24 | 128 | −0.1459 | 0.8357 |
| AWGN | −22 | 128 | −0.0888 | 0.7604 |
| AWGN | −20 | 128 | −0.0825 | 0.5045 |
| AWGN | −18 | 128 | −0.0348 | 0.3745 |
| AWGN | −16 | 128 | −0.0315 | 0.3179 |
| AWGN | −14 | 128 | −0.0486 | 0.2181 |
| Quiet | −18 | 93 | 0.0190 | 1.2872 |
| Quiet | −16 | 123 | −0.3666 | 2.0109 |
| Quiet | −14 | 124 | −0.5810 | 1.9426 |
| Moderate | −18 | 108 | −0.0323 | 0.9035 |
| Moderate | −16 | 125 | −0.1543 | 1.0760 |
| Moderate | −14 | 128 | −0.3811 | 1.3388 |
| Disturbed | −18 | 111 | −0.1535 | 0.8194 |
| Disturbed | −16 | 128 | −0.2745 | 1.0306 |
| Disturbed | −14 | 128 | −0.2657 | 0.9624 |

Table 8: SNR error relative to the configured transmit-reference level. AWGN uses known aligned tones; fading rows include only correct accepted decodes with valid estimates. All use the centered ideal frontend.

Bias is mean estimated SNR minus the configured value; RMSE is the root mean squared difference. In the unconditional AWGN rows, bias ranges from −0.1459 to −0.0315 dB and RMSE from 0.2181 to 0.8357 dB. The fading rows are conditioned on successful reception and include realized receive-power fluctuations relative to transmit-reference SNR. Their biases range from −0.5810 to 0.0190 dB and RMSE from 0.8194 to 2.0109 dB. The valid-case counts are therefore essential to interpreting these figures. No empirical bias correction is fitted.

### 6.6 Waveform duration, bandwidth, and peak power

The active waveform has exactly 116,800 samples at 12 kHz, including both filter tails, for a duration of 9.733333333 s. The physical summary uses 2304 distinct randomized source cases, counted once each despite the paired power records. Mean $B_{99}$ is 1959.236 Hz and the largest observed value is 1998.225 Hz. Mean PAPR is 0.710906 dB and the largest observed value is 0.710965 dB. These extrema belong to the sampled payloads, not to an exhaustive search over the codebook. Unit-peak active-frame mean power is approximately 0.8490 and energy 8.2636 in normalized power-seconds, compared with 1 and 9.7333 under unit-average normalization.

## 7 Discussion

The acceptance curves describe the sampled channel ensemble, receiver, frontend, and power normalization together. Across quiet, moderate and disturbed, both path separation and Doppler spread increase; the experiment changes both and cannot isolate their individual effects. Those profile names do not impose a universal ordering of frame-acceptance probability for this receiver. Curve crossings or small interpolated differences must be interpreted with the cell uncertainty and 2 dB grid resolution, not as evidence of a general ranking. Peak normalization reduces average transmit power, so the two power settings must be interpreted

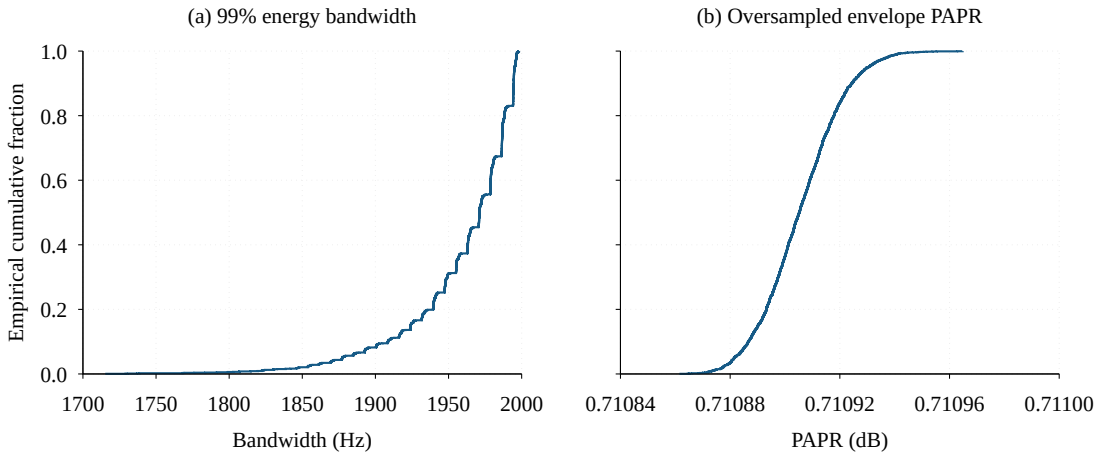



Figure 4: Empirical cumulative distributions of transmit-waveform bandwidth and PAPR over 2304 randomized source cases, without counting the paired power settings twice. Channel fading and noise do not enter these measures.

using their distinct reference-power conventions. A pooled percentage across this balanced SNR grid is not the probability of success on an arbitrary HF link.

The receiver integrates each symbol without a channel estimate. Doppler variation and delay-dependent spectral cancellation can redistribute energy within and between its tone projections. The nonbinary decoder combines these soft observations, but the fixed metric parameter $\gamma$ and the fixed pilot threshold are not adapted to the individual fading realization. The present data characterize their joint behavior; they do not identify which stage caused each erased frame.

The frontend removes the public noise guards, making the in-band estimate important to both soft decoding and final acceptance. The selected estimate can be affected by acquired noise peaks, signal leakage, and noise color. The waveform and filter geometry alone therefore do not establish a false acceptance rate; the noise-only records measure that endpoint directly. An evidence-gate exceedance without a raw decode is not an accepted message. If a noise lot contains no raw successes, it cannot demonstrate an incremental error reduction caused by gating, regardless of its evidence distribution.

The principal fading experiment samples only small CFOs. Successful tests at the full acquisition boundaries in AWGN do not establish the same probability of success at those boundaries with simultaneous dispersive fading. Physical filter transition bands, passband ripple, impulsive noise, interference, frequency drift, nonlinear amplification, and additional delay/Doppler profiles remain outside the measured models. PCM conversion is covered by software roundtrip tests, not by the floating-point channel simulation. No comparison with another modem is made.

The SNR estimate is conditioned on a recoverable payload in normal operation. Fading changes both received power and its distribution among tone projections, while success conditioning selects a subset of realizations. Consequently its error relative to the configured transmit-reference SNR is not solely instrumentation error. The measured bias is left uncorrected; the decoder does not claim to measure Doppler spread or SNDR.

# 8 Implementation and reproducibility

The standalone distribution supplies Fortran source, command-line programs, and static and shared libraries. Headerless float32 little-endian IQ stores interleaved I/Q samples at 12 kHz. The optional PCM16 mono WAV frontend uses a selectable audio center, 1500 Hz by default. The interface guide defines file bounds, result fields, the C layout, and serialized access to the shared FFT cache [8]. These interface details do not enter the statistical denominators.

The study directory contains the per-trial signal and noise records, their checksums, the channel and frontend implementation and single-trial reproduction harness, and a Fortran analysis program. Its commands regenerate the cell tables and plotting data; the plotting script renders those results without recomputing the statistics [9].

The campaign manifest records the executed source and program hashes, compiler, flags, seeds, and output identities. Exact random streams depend on the compiler/runtime. Acquisition-boundary CSVs also record CPU time, which varies across runs: their published hashes identify one execution, not a bit-identical replay requirement. Deterministic outcome comparisons exclude that timing field.

A GNU Fortran/C rebuild and the regression suite verify the implementation separately from the performance study. The suite covers core coding, waveform fixtures, file bounds and malformed inputs, automatic and aligned-input reception, the C result layout, and library roundtrips. Assertion counts are not channel trials. The fixed-seed SNR validation is a separate executable. Waveform regression checks use the complete CF32 frame and PCM samples, not only duration or peak-level summaries. The exact CF32 roundtrip checks the integer representations of quantized samples. Strict compilation treats compiler warnings as errors [10].

The code is distributed under GPL-3.0-or-later. Nico Palermo's graph, coefficient, algorithm and metric attribution and the relevant source-package identifiers are retained in `licenses/QRA-NOTICE.txt`. Checksums identify the delivered source and data.

# 9 Conclusion

QF256 transmits a 77-bit message in a 9.733333-second filtered 256-tone burst and combines blind pilot acquisition with nonbinary soft decoding and a fixed acceptance rule. CRC-valid outputs include a 2500 Hz-referenced SNR estimate evaluated with known aligned AWGN tones and accepted-correct fading messages. The study defines cell-level frame-acceptance measurements under the three ITU-R F.1487-0 mid-latitude profiles and two power constraints. The sampled unit-average crossings lie near $-19.3$ to $-19.5$ dB and unit-peak crossings near $-18.7$ dB, with overlapping conservative grid brackets. The independent noise check accepts no messages in 4096 captures; this is finite-sample evidence, not proof of zero false-acceptance probability. Acquisition-boundary and waveform measurements complement the frame-acceptance results.

# A Graph specification

The input indices $v_i$, $i = 1, \ldots, 51$, for (2)–(3) are the following sequence, read left to right and top to bottom:

```
14, 2, 4, 5, 9, 13, 10, 15, 11, 6, 1, 8, 2, 12, 9, 10, 13,
7, 4, 11, 8, 6, 3, 14, 13, 5, 9, 1, 2, 12, 3, 10, 15, 6,
7, 14, 8, 13, 12, 3, 10, 1, 11, 5, 8, 15, 9, 12, 4, 7, 11
```

The coefficient exponents $\ell_i$ are:

```
0, 14, 0, 0, 13, 37, 0, 27, 56, 62, 29, 0, 52, 34, 62, 4, 3,
22, 25, 0, 22, 0, 20, 10, 0, 43, 53, 60, 0, 0, 0, 62, 0, 5,
0, 61, 36, 31, 61, 59, 10, 0, 29, 39, 25, 18, 0, 14, 11, 50, 17
```

To construct $h_i$, first generate the powers of element 2 in GF(64) using polynomial `0x43`, and take the integer value represented by the six-bit element at exponent $\ell_i$. Embed that integer as the corresponding eight-bit polynomial element of GF(256). The numbers $\ell_i$ are therefore *not* used directly as logarithms in GF(256). The literal arrays and this conversion are preserved in the shipped coding source [7].

## References


[1] F. R. Kschischang, B. J. Frey, and H.-A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 498–519, 2001. doi:10.1109/18.910572.

[2] C. C. Watterson, J. R. Juroshek, and W. D. Bensema, "Experimental confirmation of an HF channel model," *IEEE Transactions on Communication Technology*, vol. 18, no. 6, pp. 792–803, 1970. doi:10.1109/TCOM.1970.1090438.

[3] International Telecommunication Union, Recommendation ITU-R F.1487-0, *Testing of HF modems with bandwidths of up to about 12 kHz using ionospheric channel simulators*, May 2000, Annex 3, Sections 3.1–3.3. Official recommendation.

[4] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. doi:10.1080/01621459.1927.10502953.

[5] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934. doi:10.1093/biomet/26.4.404.

[6] QF256, *Physical modem implementation*, `src/core/mfsk_iq.f90` and `src/core/waveform_metrics.f90`.

[7] QF256, *Nonbinary coding implementation*, `src/core/qra_soft.f90`.

[8] QF256, *Standalone implementation and interface documentation*, `README.md`, `docs/API.md`, `src/qf256_api.f90`, and `include/qf256.h`.

[9] QF256, *HF confirmation and noise-only study records and analysis*, October 2026, `study/`. Data, source and reproduction instructions in the accompanying distribution.

[10] QF256, *Build, regression and source-integrity records*, October 2026, `docs/VALIDATION.txt`, `docs/CORE.sha256`, and `docs/DELIVERY.sha256`.

[11] N. Palermo, `qra15_65_64_irr_e23.c` (2020) and `qracodes.c` (2016), QRA graph and decoder sources, GPL-3.0-or-later. Source package: WSJT-X 2.7.0+repack-1, Debian Sources, lib/qra/q65, retrieved October 10, 2026. Attribution and encoding references are preserved in `licenses/QRA-NOTICE.txt`.